

A NEW CATEGORY BETWEEN EXTERNAL FIXATION AND DEFINITIVE RECONSTRUCTION

Temporary Internal Stabilisation for Critical Long-Bone Defects

A modular intramedullary system designed for staged reconstruction, PMMA spacer support, and infection-conscious temporary fixation.



- TRL 4-5
- Validated
- PMMA Ready
- Modular
- HA-Ag Coating



KEY FEATURES

Modular Architecture
Two-component system adaptable to defect size and anatomy.

Mechanical-Axis Support
Intramedullary alignment for stable temporary fixation.

PMMA Spacer Compatibility
Designed to function with antibiotic-loaded spacers.

HA-Ag Antibacterial Interface
Selective coating in bone-contact zones.

DIFFERENTIATION

✗ **Current Limitations**

- ✗ External fixation dependency
- ✗ Lack of spacer-compatible systems
- ✗ Limited internal options for defects

✓ **CiTadel Advantages**

- ✓ Internal temporary stabilisation
- ✓ Designed for spacer-based workflows
- ✓ Defect-focused construct logic

CLINICAL USE CASES



Infected Critical Bone Defects

Osteomyelitis cases requiring staged debridement, antibiotic spacer support, and delayed reconstruction.



Blast & High-Energy Trauma

Complex limb injuries with segmental bone loss requiring temporary internal stabilisation.



Oncologic Segmental Resection

Post-resection defects needing stable internal fixation during staged limb salvage.

CLINICAL & ECONOMIC IMPACT



Reduced hospital stay



Faster patient recovery



Improved stability



Fewer complications

VALIDATION STATUS

TRL 4–5

Bench Testing Completed

Early Clinical Application

Prototype Validated

CITadel has undergone biomechanical testing and early clinical use.

Current TRL: 4–5 → Target: TRL 6 (clinical pilot)

PARTNERSHIP OPPORTUNITIES



Industrial Licensing

License for manufacturing and distribution within your product portfolio.



Co-Development

Joint engineering and regulatory pathway towards CE/MDR compliance.



Clinical Pilot Sites

Reference centre for structured clinical validation and pilot deployment.



Strategic Investment

Accelerate transition to clinical-scale adoption through strategic funding.



Maksym Baida, PhD

ORTHOPAEDIC TRAUMA SURGEON • DEVELOPER OF CITADEL

Over 15 years of clinical practice in traumatology and reconstructive orthopaedics. Associate Professor at Bogomolets National Medical University, Kyiv. CITadel was born from a direct clinical need – the absence of a reliable internal stabilisation option for patients requiring staged reconstruction.

Let's Start the Conversation

CITadel is open for collaboration with industry partners, clinical centres, and strategic stakeholders.

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